SENIOR HIGH SCHOOL • COMPUTER PROGRAMMING (.NET TECHNOLOGY)

CSS Gradient & Opacity

Lesson 4

Grade 11 | Quarter 2 | CPNT 2.2

LESSON

4

CSS Gradient & Opacity

Building depth and visual effects with gradients and transparency

CPNT 2.2

What You Will Be Able To Do

Unpacked Performance Standards

1. Apply CSS gradient properties to create visually appealing backgrounds.

2. Use CSS opacity to adjust the transparency of webpage elements.

3. Combine gradient and opacity effects in webpage design.

LESSON 4 • DISCUSSION

Gradient Basics

A gradient blends two or more colors smoothly, without needing an image file.

`linear-gradient()`

Blends colors along a straight line — great for banners, buttons, and backgrounds.

`radial-gradient()`

Blends colors outward from a center point, like a spotlight or glow effect.

`Direction & Angle`

Use keywords (to right, to bottom) or degrees (45deg) to set which way the gradient flows.

`Color Stops`

List multiple colors, optionally with % positions, to control exactly where each color transition happens.

LESSON 4 • DISCUSSION

Opacity & Transparency

Two different tools control see-through effects in CSS — each behaves differently.

◐

`opacity`

Fades an entire element — background AND its content/text — on a scale from 0 (invisible) to 1 (fully visible).

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`rgba() color`

Adds an alpha (transparency) value to a color. Only that color becomes see-through — text on top stays fully solid.

⇄

`opacity vs. rgba()`

opacity affects everything inside the element; rgba() lets you fade just the background while keeping text crisp.

✦

`Common Uses`

Image overlays, hover effects, disabled buttons, and readable text banners over photos.

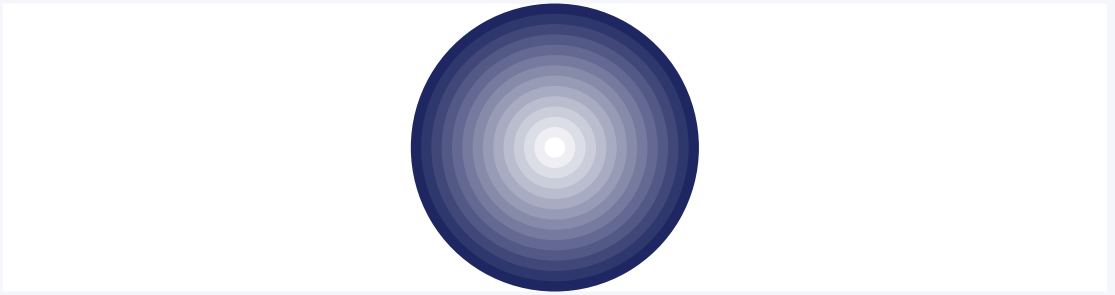
Example 1 & 2: Linear vs. Radial Gradient

Two gradient types, side by side — notice how the direction of the color blend differs.



linear-gradient(to right, navy, mint)

```
.banner {
  background: linear-gradient(
    to right, #1E2761,
    #02C39A
  );
}
```



radial-gradient(circle, white, navy)

```
.spotlight {
  background: radial-gradient(
    circle, #FFFFFF,
    #1E2761
  );
}
```

Example 3 & 4: Opacity vs. rgba()

Same navy card, two different transparency techniques — watch what happens to the text in each.



Box AND text both fade — text looks washed out.

```
.faded-card {
  background-color: #1E2761;
  opacity: 0.5;
}
```



Only the background fades — text stays fully solid.

```
.overlay-card {
  background-color: rgba(30,39,97,.5);
  color: #FFFFFF;
}
```

NEXT UP

Lesson 5: The CSS Box Model

With colors and gradients in place, we're ready to structure layouts using margin, border, and padding in Quarter 2's CPNT 2.3.

Thank you!